

TELO

Distributed Sterile-Injectable Manufacturing Under Real-World Constraints

Maximum-quality feasibility-study protocol and model specification

Central decision question

Under what product, demand, regulatory, and operating conditions can distributed regional capacity reduce sterile-injectable shortages more cost-effectively than additional safety stock, dual sourcing, reserved contract capacity, or additional centralized capacity?

The study is designed to permit a negative conclusion.

Version 1.0 | 1 September 2026 | Preliminary research design; not legal or regulatory advice

Executive decision architecture

This is not a business-plan spreadsheet and not a model built to manufacture a favorable answer. It is a pre-specified comparison of supply architectures under matched service requirements. Distributed production only wins if it is legally and technically feasible, meets the same reliability target, remains attractive after common-cause failures are modeled, and lies on the cost–reliability frontier after each alternative has been optimized fairly.

The governing rule

No strategy is eligible for a favorable conclusion unless all applicable regulatory and quality gates pass. Among eligible strategies, compare the minimum total cost required to reach the same service target—not hand-picked versions of each strategy.

What this study can establish

- A reproducible region of feasibility: combinations of utilization, release latency, fixed quality cost, supplier independence, shelf life, demand volatility, and contracted volume under which regional nodes are—or are not—efficient.
- Which bottleneck dominates for each product: transport, material lead time, testing/release, capacity, supplier concentration, contracting, or regulatory recovery time.
- Whether a distributed design remains useful when one node fails and when all nodes share an API, component, software, quality, or regulatory failure mode.
- The value of Telo's release-assurance layer as a separately gated scenario, without assuming it is validated or treating it as a substitute for sterility assurance.
- The price and take-or-pay commitment required to support resilience capacity, expressed transparently rather than hidden inside a product markup.

What it cannot establish by itself

- FDA acceptance of a particular product, site, control strategy, real-time release method, or multi-site equivalence claim.
- Manufacturing cost from CMS reimbursement or spending data; those data can be labeled utilization or pricing proxies only.
- Prospective safety or release performance from retrospective public-spectra benchmarks.
- Commercial demand without hospital, wholesaler, group-purchasing-organization, or government commitments.
- That 503B is a bridge to an approved generic pathway; it is a separate, status-dependent legal architecture.

How the Telo roadmap enters the analysis

Telo element	Role in this study	Evidence standard	Claim status
Phase 1 OS	Process orchestration, electronic records, deviation workflow, and decision support inside validated facilities	Observed workflow time, error rate, integration burden, auditability, and validated data lineage at a design-partner site	Commercial wedge; not assumed to create manufacturing capacity
Release assurance	Conditional scenario that may reduce a validated portion of batch-disposition latency	Product- and site-specific prospective validation, predefined shift response, change control, audit trail, and acceptable risk–coverage frontier	Current public-benchmark work supports methods/OOD detection only

Telo element	Role in this study	Evidence standard	Claim status
Phase 2 nodes	Distributed sterile fill-finish architecture for a tightly bounded product archetype	Qualified API, approved/contracted regulatory owner, validated process, aseptic control, stability, inspection, and executable demand contracts	Subject of feasibility model
Shelved assets	Separate future asset-development program	Asset-specific biology, IP, CMC, clinical, regulatory, and capital diligence	Outside this shortage study; do not mix into its conclusion

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1. Study charter and falsifiable hypotheses

1.1 Decision owner and decision date

Define the decision before collecting favorable data. The decision owner is the Telo research team advised by independent manufacturing, pharmacy-supply, regulatory, and operations experts. The decision is not “launch Telo.” It is whether to advance a specific product–pathway–network design into paid regulatory/engineering diligence, redesign it, or stop. Freeze the protocol, source snapshot, candidate list, and analysis version before the final simulation run.

1.2 Unit of analysis

The primary unit is a product–presentation–pathway–network architecture over a defined national or multi-region demand system. “Sodium bicarbonate” is too broad; an analyzable case is a named concentration, dosage form, container, fill volume, route, label, and regulatory pathway. Each site, supplier, component, release queue, and inventory location is modeled explicitly enough to expose shared failure modes.

1.3 Pre-specified hypotheses

Hypothesis	Falsifiable statement
H1 — proximity	Transportation distance is not usually the controlling recovery term; release latency, material lead time, and available qualified capacity dominate except in narrow disruption classes.
H2 — node economics	Regional nodes become cost-competitive only above a minimum contracted utilization and below thresholds for fixed QA cost, changeover burden, and validation replication.
H3 — pooling tradeoff	Regionalization reduces local transport and some site-concentration risk but sacrifices production and inventory pooling; the direction of net benefit changes with demand covariance and common-cause exposure.
H4 — inventory	Safety stock dominates new capacity when shelf life is long, demand is predictable, shortages are shorter than the replenishment/release cycle, and the stock is positioned before disruption.
H5 — dual sourcing	Nominal dual sourcing offers little resilience when sources share API, components, analytical constraints, ownership, geography, or regulatory failure modes.
H6 — reserved capacity	Contracted flexible capacity can dominate owned regional capacity when qualification is complete and activation/release latency is short.
H7 — release assurance	A validated release-assurance layer creates material resilience value only when it reduces an actually binding disposition delay while maintaining an acceptable error/abstention frontier under instrument, site, lot, and time shift.
H8 — product specificity	No platform-level conclusion is valid without product-specific formulation, stability, presentation, demand, and regulatory analysis.

1.4 Kill criteria

- A material regulatory or quality gate is failed or cannot be bounded credibly.
- The distributed design is dominated—higher cost and worse reliability—across the credible uncertainty region.
- Required take-or-pay volume exceeds credibly contractable demand.
- Resilience disappears under plausible common-cause dependence or when material lead time is represented correctly.

- The result depends on reimbursement being mislabeled as cost, on unsourced precision, or on release-time reductions unsupported by validated practice.
- The preferred strategy reverses under modest, plausible changes in one or two high-leverage assumptions and those inputs cannot be resolved economically.

2. Product archetype and case selection

2.1 First-node manufacturing archetype

Recommended boundary

Non-controlled, small-molecule, aqueous sterile solutions in standard vials; source qualified API and own formulation/fill-finish. Exclude biologics, vaccines, suspensions, emulsions, lyophilized products, cytotoxics, high-potency compounds, controlled substances, complex devices, and cold-chain-intensive presentations from the first model unless treated as explicit out-of-archetype comparators.

This boundary does not claim those other modalities are impossible. It keeps the first inference interpretable. Mixing sterile solutions, lyophilized products, biologics, and tablets in one economic result creates a platform claim whose average is meaningful for none of them.

2.2 Two-stage selection procedure

1. Create a dated longlist of at least five product presentations from frozen FDA and ASHP shortage snapshots, including at least one resolved/recurrent comparator.
2. Apply hard gates. A failed hard gate excludes a product from the first archetype before weighted scoring.
3. Score only surviving candidates using pre-declared weights. Keep the raw evidence, disagreements, and uncertainty—not only the final number.
4. Select one case with plausible regional relevance and one hard comparator that tests whether the conclusion generalizes.
5. Have one hospital pharmacy leader and one sterile-manufacturing expert independently challenge the selection before model calibration.

2.3 Hard product gates

Gate	Pass condition
Product identity	Exact ingredient, strength, dosage form, route, presentation, container closure, and label can be fixed.
Manufacturing fit	Aqueous small-molecule sterile solution compatible with the first process train; no hidden lyo/suspension/high-potency requirement.
Control status	Non-controlled and no special handling regime that invalidates the first-site design.
Regulatory pathway	At least one supportable approved-generic/CMO architecture can be mapped; 503B, if analyzed, is separate.
Demand observability	National/regional utilization can be bounded using hospital/GPO/wholesaler evidence, not only reimbursement.
Clinical substitutability	Shortage impact and presentation-specific substitution behavior can be described by clinicians/pharmacists.
Input feasibility	API, vial, stopper, seals, labels, and testing can be sourced and their dependencies mapped.
Economic relevance	Volume is not so large that “micro” nodes are obviously mismatched, nor so small that validation/fixed QA is necessarily overwhelming without contracted premiums.

2.4 Weighted score after gates

Criterion	Weight	Required evidence
Clinical criticality	20%	Hospital role, substitution difficulty, patient-level consequences
Shortage recurrence/duration	20%	Frequency, duration, and persistence across FDA/ASHP snapshots
Supplier vulnerability	15%	Finished-dose, API, component, ownership, and geographic concentration
Manufacturing fit	15%	Process simplicity is not assumed; score against actual first-node train
Public-data strength	10%	Labels, approvals, utilization proxies, presentations, and shortage history
Regional relevance	10%	Whether geography or response speed plausibly matters after release/material delays
Demand/contract fit	10%	Bounded demand and credible institutional purchasing model

2.5 Provisional cases—not final selections

Presentation	Provisional role	Diligence needed
Sodium bicarbonate injection 8.4%, 50 mL vial	Current/recurrent case candidate	Strong first-archetype fit; confirm presentation-level demand, current status, supplier map, container compatibility, process/sterilization route, and economics
Norepinephrine bitartrate 1 mg/mL, 4 mL vial	Resolved/recurrent comparator	Clinically important, but presentation, dilution practices, light/stability, supplier structure, and actual shortage history require reconciliation
Acyclovir sodium injection	Reserve	Retain only after confirming formulation/presentation/process fit and demand evidence
Furosemide injection	Reserve	Useful simpler comparator if data and supplier evidence are stronger
Acetazolamide sodium 500 mg vial	Exclude from first archetype	Lyophilized presentation violates the aqueous-solution boundary; useful future out-of-archetype test

Freeze the snapshot

Record the FDA/ASHP status, access date, shortage reason, available presentations, applicants/labelers, and links in a machine-readable product dossier. A dynamic webpage consulted later is not a reproducible dataset.

3. Comparator strategies and fair-comparison rules

The model contains eight architectures. The first seven answer the durable supply-design question; the eighth is a separate shortage-response pathway. Each architecture must be optimized within its own feasible design space before comparison.

ID	Architecture	What must be represented
S0	Status quo centralized	Existing approved supply architecture; no favorable assumptions about hidden spare capacity
S1	Additional safety stock	Optimize days of supply, placement, rotation, carrying cost, expiry, and shortage timing
S2	Dual source + contracts	Two qualified sources with explicit shared-cause mapping and allocated commitments

ID	Architecture	What must be represented
S3	Reserved flexible contract capacity	Qualified capacity held through reservation/take-or-pay terms; include activation, campaign, and release delay
S4	Additional centralized capacity	New/expanded large-site capacity with scale economies and concentration risk
S5	Distributed regional nodes	Multiple smaller approved/contracted nodes using conventional validated disposition
S6	Distributed + validated release assurance	S5 plus only the release components prospectively demonstrated and accepted for the product/site
S7	503B shortage response	Separate status-dependent architecture; not ranked as though legal availability were permanent

3.1 Fair-comparison requirements

- Same demand paths and disruptions: use common random numbers so strategies face the same simulated world in each paired run.
- Same service target: minimize cost subject to a pre-specified fill-rate and shortage-risk constraint; do not compare a low-cost unreliable strategy to an overbuilt reliable one.
- Same cost boundary: include annualized capital, fixed quality/operations labor, validation, testing, deviations, scrap, inventory, distribution, software/integration, compliance, and reserved-capacity payments without double counting.
- Same time horizon and discounting: include commissioning/qualification lead time where a strategy does not exist on day zero.
- Same common-cause scrutiny: nominal site count is not independent redundancy.
- Pathway gating: an ineligible strategy is excluded—not assigned a large but finite regulatory penalty that an optimizer can pay through.

3.2 Strategy optimization variables

Architecture	Variables to optimize
Safety stock	Days of supply, locations, rotation frequency, reorder point, target service level
Dual source	Allocation fraction, capacity reservation, independence constraints, inventory by source
Reserved capacity	Reserved fraction, activation threshold, minimum campaign, take-or-pay, product slate
Central expansion	Site size, spare capacity, campaign structure, stock placement
Distributed	Node count, node size, regional assignment, pooled overflow, common suppliers, safety stock, campaign schedule
503B	Eligible time windows, products/presentations, activation policy, BUD, jurisdictional reach

4. Regulatory and quality decision architecture

Build a two-path decision map reviewed by a professional with actual generic-drug/CMC and 503B experience. Label it preliminary regulatory analysis. Do not infer permissions from a founder narrative, a proposed rule, or shortage status alone.

4.1 Approved generic / contract-manufacturing path

6. Identify the application owner and exact product/application relationship in Drugs@FDA and the Orange Book.

7. Determine whether Telo would own/sponsor an application, act as a contract manufacturer for an owner, license a product, or supply a partner. State contractual and quality-unit responsibility.
8. Map each site and manufacturing step in the application; determine submission category, review/inspection needs, and implementation timing for each change or added site.
9. Define product and process development, control strategy, equipment qualification, process performance qualification, cleaning validation, analytical method transfer/validation, media fills/aseptic validation or terminal sterilization validation, container-closure integrity, stability, and continued process verification.
10. Map deviation, OOS, CAPA, change control, data integrity, batch disposition, recall, pharmacovigilance, and complaint responsibilities in the quality agreement.
11. Treat cross-site comparability and “statistical interchangeability” as claims requiring product/site evidence and agency engagement—not as automatic platform inheritance.

4.2 503B shortage-response path

- Verify facility registration, inspection/quality status, state permissions, and distribution rules.
- Determine bulk-substance eligibility and the exact effect of FDA shortage-list status at compounding, distribution, and dispensing dates.
- Analyze essentially-copy restrictions, clinical-difference exceptions, labeling, adverse-event reporting, and transition when a shortage resolves.
- Support formulation, sterility, stability/BUD, container closure, testing, and release under applicable CGMP expectations.
- Model this pathway as a state machine that can become unavailable; do not treat 503B output as an approved-generic supply source after legal predicates disappear.

4.3 Distributed establishments and advanced manufacturing

FDA's FRAME work and the Advanced Manufacturing Technologies Designation Program provide important engagement context, but they do not remove product approval, site, establishment, validation, inspection, or CGMP requirements. If citing the 2026 distributed-manufacturing-unit registration proposal, label it proposed and re-check final status before publication. The base case should not depend on a proposed rule becoming final in Telo's preferred form.

4.4 Hard-gate matrix

Gate	Pathway	Model treatment
Application/quality owner identified	Approved/CMO	Fail = architecture excluded
Site and manufacturing steps supportable in application	Approved/CMO	Adds qualified lead time; unresolved = no conclusion
Process, cleaning, analytical, and aseptic/sterilization validation	Both	Hard feasibility condition
Stability and container-closure support	Both	Caps inventory, BUD/shelf life, and location policy
503B bulk/copy/shortage predicates	503B	Dynamic on/off state
State licensure/distribution permissions	503B/network	Restricts service region
AI/release model validated in state of control	Release-assurance scenario	If absent, use conventional release only
Cross-site comparability accepted	Distributed	No fleet-level equivalence claim if absent

5. Model boundary, network topology, and data architecture

5.1 Scope and time resolution

Use a daily time step for a five-year base horizon with warm-up sufficient to stabilize inventory and production queues. Run a weekly time-step sensitivity only after confirming it reproduces daily results. The stochastic network contains demand regions, hospitals or aggregated hospital classes, finished-goods inventories, manufacturing sites, batch queues, testing/release queues, API suppliers, critical component suppliers, distribution links, and regulatory availability states.

5.2 Boundary choices to freeze

Boundary	Base definition
Geography	Four U.S. demand regions in base case; sensitivity from 3–8; region definition must follow data and distribution practice
Demand	Presentation-specific units; do not combine concentrations or packages without explicit conversion/substitutability
Production	Formulation, sterile filtration where applicable, filling, inspection, testing, release, scrap, cleaning/changeover
Upstream	Qualified API and critical components represented as suppliers with lead times, capacity, and common-cause group
Downstream	Distribution center or direct-to-hospital policy, transport time, allocation, substitution, expiry
Quality	Deviation/rejection, testing queues, release time, model abstention, state-of-control loss
Regulation	Pathway availability, site qualification, shortage-list state for 503B, and implementation delay
Economics	Societal/purchaser supply cost primary; provider operational harm may be a secondary scenario, not an invented patient-dollar value

5.3 Data hierarchy

Tier	Evidence	Examples
Tier 1	Legal/regulatory and official product/shortage records	eCFR, FDA guidance and databases, Drugs@FDA, Orange Book, DailyMed
Tier 2	Direct operational evidence	Executed contracts, quotes, plant batch history, deviation/release logs, hospital/GPO/wholesaler demand
Tier 3	Peer-reviewed empirical evidence	Recovery, resilience, inventory, and supply-network research
Tier 4	Qualified expert elicitation	Structured ranges with role, experience, confidence, rationale, and disagreement
Tier 5	Illustrative placeholders	Allowed for scaffold testing only; never cited as results

5.4 Assumption register schema

Every material parameter must store: parameter ID, definition, units, entity level, low/base/high or full distribution, dependence group, source and page/line where possible, access date, transformation code, owner, confidence, validation status, last revision, and which outputs it can reverse. Never paste a number into the final model without provenance.

6. Deterministic techno-economic model

The spreadsheet is a transparent screening and audit layer. It should expose unit mismatches, infeasible capacity, cost double counting, and high-leverage inputs before simulation. It must not be presented as the final reliability estimate.

Annual saleable capacity = sites × batches/site-year × units/batch × yield × uptime

Eq. 1. Calculate by product and site; do not use nominal nameplate capacity.

Utilization = annual demand allocated to site ÷ annual saleable capacity

Eq. 2. Keep planned utilization distinct from realized utilization after disruptions.

CRF = $r(1+r)^n \div [(1+r)^n - 1]$

Eq. 3. Use for annualized capital when a full discounted cash-flow model is not required.

Fully loaded cost/unit = total annual strategy cost ÷ saleable units delivered

Eq. 4. Report denominator explicitly; produced and delivered units differ under expiry and shortage.

Response time = material availability + setup + production + testing/release + delivery

Eq. 5. Terms can overlap only where operationally valid; model queues in simulation.

6.1 Cost ledger

Ledger	Include
Capital	Building/module, utilities, cleanroom, process equipment, fill line, isolator/RABS, inspection, laboratory, warehouse, commissioning/qualification; annualize or model cash flows
Fixed site operations	Quality unit, production staff, maintenance, metrology, validation, regulatory, IT/data integrity, security, facility overhead
Product/site launch	Development, tech transfer, methods, stability, PPQ, media fills, submissions, inspections; amortize transparently
Variable production	API, excipients, vial, stopper, seal, label, packaging, consumables, utilities, direct labor truly variable with volume
Quality/testing	Release testing, sterility/endotoxin/assay, reference standards, environmental monitoring, retain samples, deviation investigation
Failure/waste	Yield loss, rejected batches, line clearance, changeover, expiry, obsolescence, recalls if modeled
Inventory/logistics	Carrying, warehousing, cold/light controls if applicable, transport, emergency freight, redistribution
Resilience contracts	Capacity reservation, minimum purchase, take-or-pay, supplier qualification, second-source maintenance
OS/integration	Sensors, interfaces, validation, cybersecurity, model monitoring, audit storage, user training and support

6.2 Break-even calculations

Break-even contracted volume = annual fixed/resilience cost ÷ contribution margin per delivered unit

Eq. 6. Use price net of variable production, testing, distribution, and expected waste.

$$\text{Required resilience premium} = [\text{Cost}(\text{strategy}) - \text{Cost}(\text{status quo})] \div \text{committed units}$$

Eq. 7. State whether premium is paid by hospitals, GPOs, government, or another purchaser.

Compute break-even surfaces—not one point—over node count, site size, utilization, product count per node, changeovers, release time, quality staffing, and commitment share. Multi-product pooling may improve utilization but adds changeover, scheduling, validation, and shared-failure complexity; model both sides.

7. Stochastic reliability simulation

Implement a discrete-time, event-driven network simulation in Python after the spreadsheet screen. Use seeded random streams and common random numbers across strategies. Store one tidy row per run-product-strategy plus event logs for audit. A daily step is adequate initially; use an explicit queue for production and release rather than adding average lead times after the fact.

7.1 State variables

- On-hand inventory by product, region/location, age/expiry, and regulatory status.
- Outstanding demand/backorders and allocation priority by region/hospital class.
- Raw-material/component inventory and open purchase orders by qualified supplier.
- Site capacity, uptime/failure state, campaign/setup state, batch work in process, and maintenance state.
- Testing/release queue, deviation state, batch disposition, and Telo-model state of control where applicable.
- Transport links and delay states.
- Pathway state, including 503B shortage eligibility windows and newly qualified site availability.

7.2 Core daily event order

12. Advance date and expire inventory using FEFO rules.
13. Draw/advance demand and demand shocks using the run's shared random streams.
14. Advance supplier, site, common-cause, regulatory, and transportation disruption states.
15. Receive qualified materials and released batches whose lead times complete.
16. Allocate available inventory under the pre-specified policy; record served demand, backorders, unmet units, and shortage-day flags.
17. Advance setup, production, inspection, test, deviation, and release queues; sample yield and rejection consistently with site/batch effects.
18. Trigger reorder, campaign, emergency transfer, reserved-capacity activation, and 503B policies if eligible.
19. Accrue cost and write audit events; assert mass balance and nonnegative state.

7.3 Publication pseudocode

Paired simulation

for run in 1...N:

 sample uncertain structural parameters

 generate shared demand and disruption streams

 for strategy in eligible_strategies:

 reset state with strategy-specific design

 simulate warm-up + analysis horizon

 assert mass balance and log events

 write costs, service, failures, and recovery metrics

compare paired strategy outcomes; optimize designs; bootstrap uncertainty

7.4 Randomness and dependence

Use separate reproducible random streams for routine demand, demand shocks, site failures, common-cause events, supplier delays, yield, deviations, release times, and transport. Within each run, strategies receive the same exogenous streams wherever conceptually comparable. Represent dependency with hierarchical effects or explicit cause groups rather than an arbitrary correlation coefficient alone. Example: two sites may be physically independent but share the same API source, stopper supplier, analytical method defect, OS version, or regulatory action.

8. Demand, disruption, inventory, and allocation submodels

8.1 Demand

Estimate annual presentation-specific demand and regional shares using the strongest available direct evidence. Model routine nonnegative variability with a hierarchical time-series distribution that reproduces mean, coefficient of variation, seasonality, and autocorrelation. Add compound demand shocks only when empirically or expert-supported. Separate true utilization from purchases distorted by stockpiling or allocation during shortages.

- Base case: hierarchical lognormal or negative-binomial daily/weekly demand with region and season effects.
- Shock arrivals: Poisson or empirically resampled episodes; magnitude and duration jointly distributed.
- Substitution: presentation/clinical substitution matrix reviewed by pharmacists; do not assume one vial equals another merely by ingredient name.
- Contract demand: committed volume is distinct from potential demand and determines economic utilization.

8.2 Supply disruptions

Class	Examples	Parameters
Site-specific	Equipment failure, contamination investigation, maintenance, local utility/weather event	Frequency, duration, residual capacity, restart/requalification
Quality	Deviation, OOS, sterility concern, data-integrity issue, batch rejection	Batch/site dependence, investigation duration, release blockage
Supplier	API, vial, stopper, seal, testing material, single-use component	Lead time, allocation, qualification, shared supplier graph

Class	Examples	Parameters
Common cause	Shared supplier, method, software, cyber, regulatory, ownership, geography	Affected set and conditional impact—not independent site draws
Demand surge	Clinical event, substitution from another shortage, policy change	Correlated regions, magnitude, duration
Regulatory/pathway	Site/action delay, shortage-list resolution, new-site qualification	State transition and implementation lag

8.3 Inventory and allocation

Track inventory by age and location using first-expire-first-out. A strategy cannot claim the reliability of six months of stock without paying for working capital, warehousing, rotation, and expiry, and without testing whether the stock existed before the disruption. Allocate scarce supply with a pre-specified policy—proportional, criticality-weighted, minimum-guarantee, or optimization-based—and report regional equity. The preferred result should not depend silently on starving one region.

$$\text{Inventory}[t+1] = \text{Inventory}[t] + \text{released receipts} - \text{shipments} - \text{expiry} - \text{quality holds}$$

Eq. 8. Compute by product, presentation, location, and age cohort.

$$\text{Backlog}[t+1] = \max(0, \text{Backlog}[t] + \text{Demand}[t] - \text{fulfilled}[t])$$

Eq. 9. Also report demand permanently lost or clinically substituted if modeled.

8.4 Recovery

Define recovery as a sustained return above the service threshold for a minimum window (for example, 14 or 30 days), not the first day inventory becomes positive. Report time to first response, time to stable recovery, area under the unmet-demand curve, and probability that demand is met throughout an episode. This distinguishes rapid but insufficient bursts from durable recovery.

9. Telo release-assurance submodel

Current evidence boundary

The public-spectra work is a methods proof that a conventional calibration can fail under instrument shift and that a conservative layer can detect out-of-distribution conditions and abstain. It is not yet evidence of accurate commercial batch release, target-drug chemistry, validated real-time release, sterility assurance, or useful throughput at an acceptable abstention rate.

The prior benchmark should be reported with both risk and coverage. In the cited older experiment, low unconditional wrong-release exposure was achieved primarily by abstaining on nearly all shifted cases; conditional coverage among the few released cases remained poor. That result supports stop-the-line detection, not a claim that release has been automated. The next evidence milestone is a useful risk-coverage frontier under realistic shifts, followed by prospective product/site validation.

9.1 Separate four scenarios

Scenario	State	Model treatment
R0	Conventional release	All standard validated testing and human disposition; no Telo benefit
R1	Documentation/orchestration	Reduces administrative cycle time only where measured; no analytical test replacement
R2	Shadow-mode decision support	Predictions and shift alerts do not control release; value measured through detection, investigation prioritization, and prospective concordance
R3	Validated release component	Only after predefined product/site validation and quality/regulatory acceptance; model the exact component and fallback path

9.2 Metrics that must travel together

- Released fraction (coverage) and abstention fraction.
- Wrong-release probability among all units and conditional on release.
- False-hold probability and operational cost of unnecessary abstention.
- Calibration/coverage overall and by instrument, site, lot, time, concentration range, operator, and shift type.
- Detection delay, repair sample count, and performance after repair without leakage.
- Release-time distribution including fallback testing, investigation, and revalidation—not only model inference time.

9.3 Required next benchmark design

20. Pre-register datasets, shifts, endpoints, thresholds, and failure definitions.
21. Use grouped external validation by instrument/site/lot/time; prevent spectra from related samples leaking across train/test partitions.
22. Compare against calibration transfer, standardization, domain adaptation, conformal/uncertainty baselines, and a mandatory-hold policy.
23. Evaluate selective prediction curves: risk versus coverage and operational cost versus coverage.
24. Stress test unknown shift types, adversarial sensor faults, missing metadata, label noise, drift, and repair with few verified assays.
25. Release code, seeds, data provenance, model card, negative results, and limitations; then repeat on target chemistry with a qualified laboratory partner.

10. Optimization, outputs, and decision rules

10.1 Primary optimization

minimize $E[\text{annual total cost}]$ subject to fill rate $\geq \tau$, $P(\text{fill rate} \geq \tau) \geq q$, and every hard gate = PASS

Eq. 10. Optimize each strategy separately over its allowed design variables.

Choose τ and q before viewing final results—for example, a mean fill-rate threshold plus a tail-reliability constraint. Also bound maximum shortage days, maximum regional disparity, and recovery time where stakeholders consider those independently important. A strategy is “best” only relative to explicit weights or constraints; otherwise present the Pareto frontier.

10.2 Required outputs

Metric	Definition	Report
Cost per delivered unit	Total annual cost / delivered nonexpired units	Mean, median, p5/p95
Fill rate	Fulfilled demand / demand	Overall and by region; mean and tail
Shortage days	Days daily fill falls below frozen threshold	Expected, p95, probability of zero
Unmet units	Demand not fulfilled within allowed window	Area and peak
Recovery time	Days from disruption to sustained service recovery	Median/p95 by event class
Probability meeting demand	Runs satisfying all service constraints	With Monte Carlo interval
Break-even volume	Committed units supporting fixed/resilience cost	Surface over price/utilization/node count
Incremental cost-effectiveness	Incremental cost / shortage-day avoided	Exclude dominated strategies; uncertainty interval
Regional equity	Max-min or dispersion in region-level fill/recovery	Report allocation-policy sensitivity
Quality burden	Holds, abstentions, deviations, rejected batches	Counts, duration, cost

$$\text{ICER} = [\text{Cost}(\text{strategy}) - \text{Cost}(\text{reference})] \div [\text{Shortage-days}(\text{reference}) - \text{Shortage-days}(\text{strategy})]$$

Eq. 11. Do not compute for a strategy that is more costly and less effective; label dominance.

10.3 Decision classification

Decision	Pre-specified interpretation
Advance	All gates pass; distributed design is Pareto-efficient; benefit persists through credible uncertainty; contractable demand clears break-even; no unresolvable common-cause fragility
Narrow advance	Feasible only for a defined product/presentation/region/contract structure; state the boundary as the principal finding
Redesign	A resolvable bottleneck—supplier independence, release process, site size, stock placement, or contract—causes failure
Stop	Dominated, legally/technically infeasible, dependent on unsupported release claims, or economically incompatible with credible demand

11. Uncertainty, sensitivity, and value of information

11.1 Parameter uncertainty

Represent material uncertainty as distributions or empirically resampled data, not decorative low/base/high bars. Use bounded distributions that respect physical support; preserve posterior or elicited dependence. Separate aleatory variation (events/demand) from epistemic uncertainty (unknown failure rates, costs, or regulatory timing) so the team can see what more evidence can reduce.

11.2 Mandatory sensitivity set

Input	Decision channel
Capacity utilization	Fixed cost/unit, surge headroom, break-even commitment
Release time	Whether local production can respond before stock/contract alternatives
Fixed QA/labor per node	Penalty for replicating qualified quality systems
Yield and uptime	Effective capacity and downside tail
Demand volatility/covariance	Pooling benefit versus regional responsiveness
API/component lead time	Whether local fill-finish is upstream-constrained
Common-cause dependence	Whether node count is real redundancy
Shelf life/expiry	Inventory option value and distributed inventory waste
Node count/size	Scale-redundancy frontier
Contract commitment	Economic utilization and purchaser premium
Deviation/rejection and release assurance	Quality-driven delay, abstention, and false-hold burden

11.3 Analysis methods

- One-way thresholds for communication and debugging.
- Global variance-based sensitivity (Sobol where computationally feasible) or partial rank correlation for monotone relationships.
- Structural alternatives: demand distribution, failure process, allocation rule, recovery definition, and site-dependence representation.
- Decision-reversal map: show where the preferred architecture changes, not only output variance.
- Expected value of perfect/partial information for high-leverage unknowns to prioritize interviews, quotes, data access, or pilots.

11.4 Decision-reversal reporting

The strongest figure

A two- or three-dimensional feasibility map showing which strategy minimizes cost at the service target across capacity utilization, release time, and common-cause dependence—with infeasible regions masked by regulatory/quality gates. This communicates more than a single optimistic base-case bar chart.

12. Validation, interviews, and independent review

12.1 Five-layer validation

Layer	Required work
Conceptual	Manufacturing, pharmacy, regulatory, and OR experts review boundaries, causal logic, comparators, and failure modes

Layer	Required work
Input	Source reconciliation, direct quotes/records where authorized, structured elicitation, range/confidence review
Computational	Unit tests, mass balance, extreme cases, seed reproducibility, independent code walkthrough
Empirical/backcast	Recreate selected historical shortage episodes without fitting the desired strategy ranking
Decision	Blind or pre-specified review of whether conclusions follow from gates, uncertainty, and dominance

12.2 Historical backcasting

Select multiple shortage episodes beyond the two focal products, including at least one site failure, one upstream/common-cause disruption, one demand shock/substitution cascade, and one resolved shortage. Freeze the information available at episode start, simulate candidate architectures, and compare broad duration, severity, recovery shape, and binding constraints. Backcasting validates behavior, not counterfactual proof. Document mismatches and revise openly.

12.3 Interview program

Role	Target	Purpose
Hospital pharmacy / supply chain	6–8	Shortage workflow, allocation, substitutions, inventory, contracting, data
Sterile manufacturing / process engineering	5–6	Batch/campaign constraints, OEE, changeover, scale, startup/recovery
Quality / CMC / microbiology	4–5	Testing, release, deviations, validation, sterility assurance, costs
Generic-drug / 503B regulatory	3–4	Pathway gates, site changes, shortage transitions, cross-site claims
GPO / wholesaler / distributor	2–3	Demand visibility, contracting, allocation, transport, resilience premiums
Drug-supply economics / operations research	2–3	Comparator optimization, dependence, calibration, welfare boundary

Target 25–30 completed conversations if access permits. Count a conversation only when the interviewee has directly relevant experience and the evidence is logged. Use a standard core guide plus role-specific modules. Obtain consent for attribution; otherwise record role and organization type anonymously. Do not describe interviewees as partners, validators, or advisors without authorization.

12.4 Independent review standard

- Reviewer 1: sterile-injectable manufacturing/quality professional who can reject process and cost assumptions.
- Reviewer 2: generic-drug/CMC or 503B regulatory professional who can reject the pathway map.
- Reviewer 3: operations-research/simulation expert who can reject the model structure, optimization, and uncertainty methods.
- Reviewer 4 (preferred): hospital pharmacy/GPO/wholesaler leader who can reject demand, inventory, and contract assumptions.
- For each review, publish the issue, decision, exact model change, and rationale. A logo or flattering quote is not validation.

13. Exact execution sequence and publication package

13.1 Work sequence

26. Version and timestamp this protocol; pre-specify service thresholds, decision classes, and kill criteria.
27. Freeze FDA/ASHP/Drugs@FDA/Orange Book/DailyMed snapshots and create five machine-readable product dossiers.
28. Apply hard gates and weighted scoring; have two domain experts challenge the case selection.
29. Commission preliminary approved-generic/CMO and 503B regulatory maps; mark every uncertainty as a gate.
30. Build a supplier/component/site dependency graph for both product cases.
31. Collect direct demand, operations, cost, release, stability, and contract evidence; replace yellow workbook cells.
32. Run the deterministic model and resolve every mass-balance, capacity, annualization, and source check.
33. Implement the tested Python simulation with common random numbers and event logs.
34. Optimize every strategy to matched service constraints; then run global and structural sensitivity analyses.
35. Backcast multiple historical episodes; reconcile mismatches and freeze the analysis version.
36. Obtain at least three independent expert reviews; publish the revision log.
37. Release the paper, executive summary, workbook, code, assumptions, data dictionary, model card, tests, review record, and limitations together.

13.2 Repository structure

Recommended public repository

```

README.md
protocol/protocol_v1.pdf
data/raw_snapshots/ data/processed/ data_dictionary.csv
model/deterministic.xlsx
src/demand.py disruptions.py production.py quality.py allocation.py economics.py optimize.py
tests/test_mass_balance.py test_extremes.py test_reproducibility.py
config/product_A.yaml product_B.yaml strategies.yaml
results/frozen_run_manifest.json summary.csv figures/
reports/executive_summary.pdf paper.pdf assumptions.xlsx reviewer_revision_log.csv
LICENSE CITATION.cff CHANGELOG.md

```

13.3 Publication-grade outputs

- A 15–25 page paper with methods sufficient to reproduce the result, plus appendices.
- A two-page executive decision memo that states the decision region, dominant bottlenecks, and conditions that reverse the result.
- A public workbook with no unsourced material input and a visible readiness/check sheet.
- Versioned Python code, environment lock, seeds, run manifest, tests, and tidy simulation output.
- Product dossiers, regulatory decision maps, assumption/source register, interview synthesis, and revision log.
- At least one non-obvious finding and one explicit negative or null result; no obligation to conclude distributed wins.

13.4 Recognition and distribution

Distribute for criticism before publicity. Send the protocol or targeted result—not a generic pitch—to professionals who can invalidate it: hospital pharmacy-supply leaders, sterile-manufacturing/quality executives, generic-drug

regulatory experts, wholesalers/GPOs, and operations researchers. After revisions, submit the research to an appropriate peer-reviewed operations, health-systems, pharmaceutical-quality, or supply-chain venue and archive code/data with a DOI where licensing permits. A strong public artifact plus documented expert criticism is more credible than mass cold-email endorsements.

13.5 College-application use

Application positioning

The value is not that the report was created “for M&T.” It is evidence that you interrogated your own venture with engineering, regulatory, and commercial discipline. List the work as one Telo activity or project, quantify only completed evidence, and use the most surprising real conclusion. In an essay, explain how evidence changed a design decision; do not summarize the model or force Penn names into it.

13.6 Claim discipline

Do not say	Use only after evidence exists
“FDA-ready” or “regulator validated”	“Preliminary regulatory map reviewed by [qualified role]; product/site engagement still required”
“Our AI automates release”	“In a public-benchmark methods study, our shift detector abstained under instrument drift; prospective product/site validation is the next step”
“Distributed manufacturing solves shortages”	“The model found a bounded feasibility region—or found distributed capacity dominated—under stated assumptions”
“Partnered with” after an interview	“Interviewed [role/organization type] and revised X assumption”
“Manufacturing cost” from CMS	“CMS utilization/reimbursement proxy; cost estimated separately”

Appendix A. Core variable dictionary

Symbol	Definition	Unit/state
p, r, s, k, t	Product, region, site, supplier/component, time	indices
D[p,r,t]	Demand for presentation p in region r at time t	units/day
I[p,r,a,t]	Released inventory of age cohort a	units
RM[p,k,t]	Qualified raw-material/component inventory	units or kg
Q[p,s,t]	Nominal production capacity	units/day
U[s,t]	Site uptime/capacity fraction after events	fraction
Y[p,s,b]	Saleable batch yield	fraction
B[p,s,b]	Batch size	units
Lmat[p,k]	Material procurement lead time	days
Lprod[p,s]	Setup plus production/inspection time	days
Lrel[p,s,b]	Testing, investigation, and disposition time	days
Lship[s,r]	Shipment time from site/location to region	days
Fail[s,t]	Site-specific failure state	binary/fraction
Common[g,t]	Common-cause group event state	binary/fraction
Reject[p,s,b]	Batch rejection state	binary
Arel[p,s,b]	Release-assurance abstention state	binary
Gate[j]	Regulatory/quality gate j	PASS/FAIL/UNCERTAIN
Ccap, Cfix	Annualized capital and fixed operations/quality cost	\$/year
Cvar, Ctest	Variable production and testing cost	\$/unit; \$/batch
Cinv, Cdist	Inventory and distribution cost	\$/year; \$/unit
Cres	Capacity-reservation/take-or-pay cost	\$/year
FR	Demand fill rate	fraction
SD	Shortage days under frozen daily threshold	days/year
RT	Time to sustained recovery	days/event

Appendix B. Equation and reporting library

Fill rate = Σ fulfilled demand $\div$ Σ demand

B1. Report overall, by region, and lower-tail across runs.

Shortage-day indicator[t] = $1\{\text{fulfilled}[t] \div \text{demand}[t] < \tau_{\text{day}}\}$

B2. Freeze τ_{day} before analysis.

Expected shortage days = mean_runs Σ_t indicator[t]

B3. Also report p95 and probability of zero shortage days.

Probability meeting demand = mean_runs 1{all pre-specified service constraints pass}

B4. Use a Monte Carlo confidence interval.

Regional disparity = max_r(metric[r]) – min_r(metric[r])

B5. Choose fill shortfall or recovery time and state direction.

Expiry rate = expired units ÷ units released into inventory

B6. Report by strategy and location.

Risk among releases = wrong releases ÷ released predictions

B7. Always pair with released fraction.

Unconditional wrong-release exposure = wrong releases ÷ all evaluated cases

B8. Low exposure can be caused by near-total abstention.

Monte Carlo standard error = sample SD ÷ $\sqrt{N}$

B9. Increase N until decision-relevant contrasts stabilize.

Paired effect = mean_run [Outcome(strategy) – Outcome(reference)]

B10. Common random numbers reduce comparison noise.

Appendix C. Computational QA and falsification tests

Test	Expected behavior
Mass balance	Beginning inventory + receipts = shipments + expiry + holds/rejections + ending inventory within tolerance
Nonnegativity	No negative inventory, demand, time, probability, capacity, or cost
Zero demand	No shortage and no variable production; fixed costs remain per strategy
Infinite supply	Fill rate approaches 1; shortage days approach 0
Zero capacity	No new production; results depend only on initial stock/substitution
No failures	Stochastic model reproduces deterministic throughput within sampling tolerance
Perfectly correlated sites	Nominal extra sites provide no independent site-failure resilience
Independent sites	Reliability improves in the expected direction as redundancy increases
Zero shelf life	Stockpiling becomes infeasible/high waste
Infinite shelf life	Expiry disappears; carrying cost remains
Release assurance off	R1–R3 reduce exactly to conventional scenario when benefit parameters are zero
Total abstention	No wrong automated release but no release-time benefit; false-hold/fallback burden appears
Seed reproducibility	Identical config + seed produces identical outputs and event log hashes
Common random numbers	Paired strategies receive identical exogenous events by run

Test	Expected behavior
Optimization sanity	Relaxing a constraint cannot worsen the optimal objective unless numerical tolerance explains it

Appendix D. Interview guide

Core questions for every qualified interview

38. During a shortage, what actually prevents supply from responding first, and what is commonly blamed incorrectly?
39. Which assumption in this model sounds least realistic? What evidence or range would you use instead?
40. Under what exact product or event would regional production help—and under what conditions would it be irrelevant?
41. Which dependencies make apparently separate sites or suppliers fail together?
42. What decision would you make differently if this model's conclusion were true? What evidence would you require?
43. What cost, delay, or compliance activity do outsiders routinely omit?

Role-specific modules

Role	Probe
Hospital/GPO	Days of supply, allocation, substitution, emergency sourcing, contract commitments, premium willingness, data availability
Manufacturing	Batch/campaign scale, OEE, changeover, line constraints, startup after deviation, fixed staffing, multi-product pooling
Quality/CMC	Release-time decomposition, sterility/endotoxin/assay, deviations, method transfer, validation replication, acceptable AI role
Regulatory	Application/site architecture, submission timing, 503B transitions, DME implications, cross-site comparability, claim wording
Distributor	Inventory location, transport contribution, allocation rules, demand signal distortion, emergency logistics
Operations research	Failure dependence, demand process, comparator optimization, calibration, outcome thresholds, dominance/VOI

Evidence log after every interview

- Anonymous interview ID, date, role category, organization type, and relevant experience.
- Claim or assumption challenged, evidence/range offered, confidence and rationale.
- Exact model/code/source-register change—or reason for no change.
- Consent and attribution status; follow-up question and contradiction with other evidence.

Appendix E. Core source register

- **S01 — FDA Drug Shortages.** Official current/resolved status and stated reasons; freeze a dated snapshot. [Source](#)
- **S02 — FDA Drug Shortages Task Force report.** Root causes and historical shortage characteristics. [Source](#)
- **S03 — ASPE: Preventing shortages and supply-chain vulnerabilities.** Policy and market-design context for generic sterile injectables. [Source](#)
- **S04 — GAO-25-107110: Drug shortages.** Recent federal evidence on shortage trends, duration, and coordination. [Source](#)

- **S05 — GAO-25-107153: Nonprofit pharmaceutical organizations.** Comparator evidence on contracts, inventory, dual sourcing, and Civica's experience. [Source](#)
- **S06 — 21 CFR Part 211.** CGMP legal requirements; obtain qualified interpretation. [Source](#)
- **S07 — FDA aseptic-processing guidance.** Sterile drug products produced by aseptic processing. [Source](#)
- **S08 — FDA process-validation guidance.** Process-validation lifecycle and state of control. [Source](#)
- **S09 — ICH Q8(R2) Pharmaceutical Development.** Pharmaceutical development and real-time release-testing context. [Source](#)
- **S10 — FDA FRAME initiative.** Advanced/distributed manufacturing regulatory-policy context. [Source](#)
- **S11 — 2026 proposed DME registration/listing rule.** Proposed rule only; re-check final status. [Source](#)
- **S12 — FDA AMT Designation Program.** Advanced-manufacturing engagement/designation; not product approval. [Source](#)
- **S13 — FDA: Compounding when drugs are on the shortage list.** Shortage-list exceptions and transitions. [Source](#)
- **S14 — FDA: Bulk substances used under section 503B.** Bulk-substance eligibility gate. [Source](#)
- **S15 — FDA essentially-copy guidance.** Essentially-copy restrictions and shortage exceptions. [Source](#)
- **S16 — Drugs@FDA data files.** Applications, products, sponsors, and regulatory records. [Source](#)
- **S17 — Orange Book data files.** Approved products, applicants, and reference-product context. [Source](#)
- **S18 — DailyMed.** Submitted labels, formulation/presentation, storage, and packaging. [Source](#)
- **S19 — CMS Part B spending by drug.** Carefully labeled utilization/pricing proxy only. [Source](#)
- **S20 — CMS ASP pricing files.** Reimbursement/ASP proxy; never manufacturing cost. [Source](#)
- **S21 — ASHP current drug shortages.** Candidate discovery and presentation-level availability; criteria differ from FDA. [Source](#)
- **S22 — DailyMed sodium bicarbonate 8.4% label.** Provisional Product A presentation reference. [Source](#)
- **S23 — DailyMed norepinephrine label.** Provisional Product B presentation reference. [Source](#)
- **S24 — Lückert et al., Omega.** Pharmaceutical supply-chain resilience and mitigation framing. [Source](#)
- **S25 — Lee et al., Management Science.** Learning and recovery in drug shortages; validate applicability and data access. [Source](#)

Appendix F. Definition of finished

Area	Completion test
Protocol	Versioned before final analysis; hypotheses, thresholds, kill criteria, and decision rules frozen
Cases	Two exact product presentations selected from a five-product dated screen; gates and dossiers complete
Comparators	All eight architectures represented; each durable strategy optimized to the same service target
Regulation	Approved/CMO and 503B maps separated; applicable gates reviewed by qualified expert
Inputs	No material illustrative assumptions remain; distributions, units, sources, dates, and confidence recorded
Model	Deterministic screen, tested stochastic network simulation, common random numbers, event logs, and optimization
Uncertainty	Global and structural sensitivity, decision-reversal map, and value-of-information analysis
Validation	Mass-balance/extreme tests, historical backcasts, 25–30 relevant interviews or justified saturation, and ≥ 3 independent reviews
Transparency	Public code/workbook, product dossiers, source register, assumptions, run manifest, revision log, and limitations
Finding	At least one non-obvious result; conclusion may be narrow, negative, or conditional
Claims	Every external sentence maps to completed evidence; no partner/regulatory/release overstatement

Final standard

The study is application-worthy and externally respectable when a skeptical expert can trace every conclusion from a frozen question through sourced inputs, hard regulatory gates, tested code, fair comparator optimization, uncertainty, criticism, and revision—and could reproduce the result even if it is unfavorable to Telo.